

Experimental Determination of CO₂ Solubility in Liquid CH₄/N₂ Mixtures at Cryogenic Temperatures

Ting Gao, Taotao Shen, Wensheng Lin,* Anzhong Gu, and Yonglin Ju

Institute of Refrigeration and Cryogenics, Shanghai Jiao Tong University, Shanghai 200240, China

ABSTRACT: Coalbed methane (CBM) is an atypical natural gas that may contain a large amount of nitrogen in addition to methane. When liquefying such CH₄/N₂ mixtures, knowledge of solid solubility is very important and, in particular, CO₂ solubility data are essential for determining the CO₂ purification index. Although experimental data for CO₂ solubility in pure methane are available for various conditions, such experimental data for CH₄/N₂ mixtures are rare. In this paper, a solid–liquid equilibrium experimental apparatus using a static-analytic method is described. Data for CO₂ solubility in pure liquid methane were obtained and their accuracy was verified by comparison with published values. Solubility of CO₂ in CH₄/N₂ mixtures with nitrogen molar fractions of 10, 30, 50, and 70% were then measured at cryogenic temperatures between −190 and −150 °C.

1. INTRODUCTION

Solubility of impurities plays a key role in gas liquefaction: If the concentration of an impurity exceeds its solubility limit, solid precipitation occurs, leading to potential blockage of process equipment and piping.¹ In the liquefied natural gas (LNG) industry, CO₂ is a typical impurity that can precipitate at cryogenic working temperatures and cause obstructions in a plant, so a CO₂ purification process is essential for natural gas liquefaction.

The CO₂ purification index in conventional LNG technology has been standardized as 50 ppm, based on the existing solubility data of CO₂ in pure methane, because methane is the major component of conventional natural gas.² The composition of unconventional natural gas resources may differ from conventional natural gas and in recent years, they have attracted significant attention. Coalbed methane (CBM) is one of the most valuable unconventional natural gases, and it is often extracted after mixing with air. Consequently, the CBM contains a large amount of O₂ and N₂. While O₂ must be removed before liquefaction for the sake of safety, N₂ can be separated via distillation after liquefaction. In this way, the mixture to be cooled and liquefied would be primarily composed of CH₄ and N₂ (with a nitrogen content as high as 70%).^{3,4} At the same cryogenic temperature, CO₂ solubility in CH₄/N₂ mixtures may be different from that in the pure CH₄. Furthermore, due to the high proportion of N₂, liquefaction temperature would be much lower for this kind of CBM than for ordinary natural gas, which will also result in a large difference in the CO₂ solubility during the liquefaction process. As a result, the standard CO₂ purification index used in conventional LNG technology may not be applicable to CBM liquefaction so solubility data for CO₂ in CH₄/N₂ mixtures at cryogenic temperatures are required to guide the determination of the CO₂ purification index for CBM. Unfortunately, this kind of data has been unavailable to date.

There have been experimental data for CO₂ solubility in methane since the 1950s. In 1954, Donnelly and Katz⁵ measured solid–liquid–gas points for the binary system of CO₂ and CH₄ at −78 to −58 °C. In 1959, Pikaar et al.⁶ carried out solid–liquid equilibrium (SLE) experiments for a CO₂/CH₄ binary system at isothermal lines for a number of temperature points between

−160 and −80 °C. In 1962, Davis et al.⁷ obtained solubility data for CO₂ in CH₄ at temperatures between −143.5 and −72 °C using the static analytic method, which covers the widest temperature range and is the most widely referenced.

Recently, theoretical data for CO₂ solubility in CH₄/N₂ mixtures have been published. Shen et al.⁸ calculated CO₂ solubility in saturated liquid methane by the regular solution method as well as the equations-of-state (EOS) method (both Peng–Robinson (PR)⁹ and Soave–Redlich–Kwong (SRK)). This confirmed that the EOS method agrees better with the experiment data from Davis et al.,⁷ and so the PR EOS was further applied to the calculation of solubility of CO₂ in CH₄/N₂ mixtures, with nitrogen molar fractions from 0.5 to 20%. Zhang et al.¹⁰ simulated CO₂ solubility in LNG with Neqsim software, using the SRK EOS for the fluid phase and a fugacity model for the solid phase. The natural gas they studied contained 5.9 mol % nitrogen and the results also showed good agreement with experimental data.

Since 1950, solubility measurements have been carried out using different operating modes such as filtering off a saturated solution sample¹¹ or using a spectroscopic analysis of saturated solutions.¹² More recently, the static analytic method with online gas chromatographic analyses has been more widely used to obtain accurate experimental data.^{13–15} Based on this method, a SLE experimental apparatus was designed, constructed, and tested in the current work. Data for CO₂ solubility in pure liquid methane were first obtained using this apparatus and compared with both experimental data from Davis et al.⁷ and calculated data from Shen et al.⁸ to verify the accuracy of this apparatus. Solubility data for CO₂ in CH₄/N₂ mixtures were subsequently measured at cryogenic temperatures.

2. EXPERIMENTAL APPARATUS

2.1. Apparatus Sketch. The SLE experimental apparatus used in the present work is shown schematically in Figure 1,

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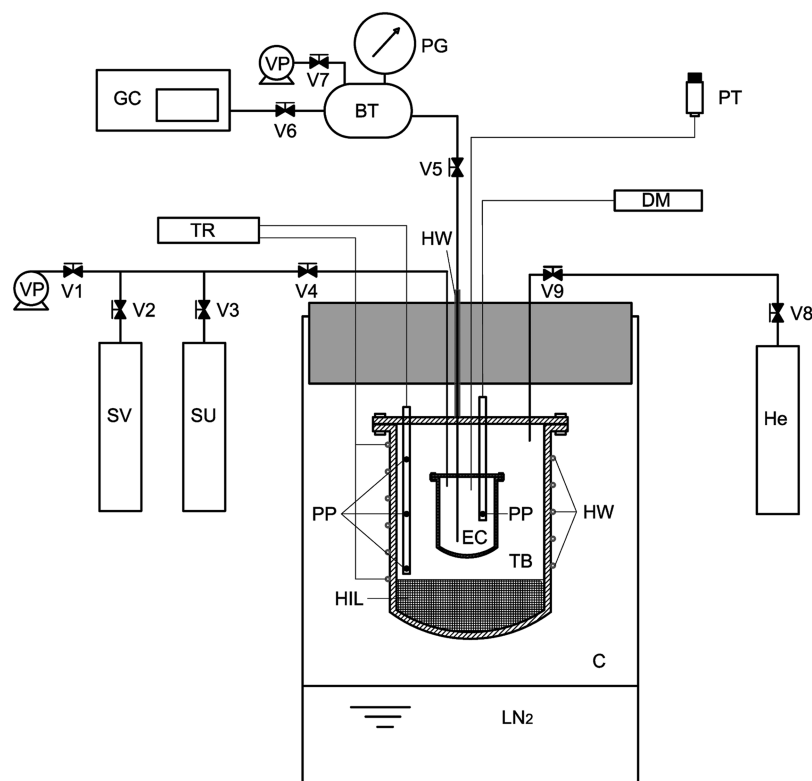


Figure 1. Static-analytic apparatus for SLE measurements.

BT: buffer tank; C: cryostat; DM: digital multimeter; EC: equilibrium cell; GC: gas chromatograph; He: helium reserve; HIL: heat insulation layer; HW: heating wire; PG: pressure gauge; PP: platinum probe; PT: pressure transducer; SU: solute reserve; SV: solvent reserve; TB: thermostatic bath; TR: PID temperature regulator; Vi: valve i; VP: vacuum pump

whereby the method was based on a static analytic method with gas chromatographic analyses for liquid samples. The principle of the static method is very simple: a solute–solvent mixture with excessive solvent was enclosed in a cell and allowed to stand for as long as required to reach solid–liquid equilibrium. The saturated liquid was then sampled and sent to a gas chromatograph for analysis. The apparatus mainly consisted of two parts: (1) a constant temperature system, and (2) a measurement system.

2.2. Constant Temperature System. The constant temperature system included an equilibrium cell (EC), a thermostatic bath (TB), a cryostat (C), a heating source (HW), and a temperature control system (TR), as shown in Figure 1.

The equilibrium cell was made of stainless steel, with a maximum pressure resistance of 6 MPa. It was important to ensure that the sample volume was small enough relative to the total liquid volume in the equilibrium cell that the sampling procedure would not perturb the equilibrium. Thus, the volume of the equilibrium cell was determined to be 350 mL (68 mm internal diameter and 100 mm high), which is about 1000 times each sampling volume (200 mL of gas volume needs only about 0.3 mL of liquid). There was an injection tube and a sampling tube in the equilibrium cell. The former was connected to the solute and solvent reservoirs, controlled independently by valves V2 and V3, and the latter was connected to the gas chromatograph through a buffer tank to reduce the sample pressure below 0.6 MPa (gauge pressure). The sampling tube was thin enough, with an inner diameter of 1 mm, to ensure the sampling procedure would not perturb the equilibrium. In addition, the sampling tube was partially wrapped with heating wires (HW) (with a heating power of 20 W) from the top of the thermostatic bath to the top of the cryostat to avoid formation of solid CO₂ plugs during sampling.

A thermostatic bath was used to provide a constant temperature environment, suspended in the cryostat, also made of stainless steel, with a volume of about 5.5 L (153 mm internal diameter and 300 mm high). To reduce the vertical temperature gradient, heat insulation materials (HIL) were attached to the bottom of the bath. Furthermore, helium could be added to the bath to improve heat transfer and consequently accelerate equilibrium.

The cooling energy was provided by a liquid nitrogen biological container (C) with a volume of 100 L, partially filled with liquid nitrogen (about 20 L), and the thermostatic bath was cooled by the vapor from the liquid nitrogen. The liquid level was measured manually periodically and the liquid nitrogen was supplemented as needed. The consumption of liquid nitrogen was about 5–10 L/d.

The temperatures we studied were higher than the temperature that liquid nitrogen can provide, so a heating source was required to attain the working temperature. A fiberglass heating belt filled with heating wires (HW) was wrapped evenly around the outside walls of the thermostatic bath. The maximum heating power of the heating belt was 80 W and the power could be adjusted by a temperature control system.

Temperature of the thermostatic bath was controlled by a proportional-integral-differential (PID) temperature regulator (TR) (AIJ, Xiamen YuDIAN Automation Technology Co., Ltd., Xiamen, China). By adjusting the output power of the heating belt, the temperature at the midheight of the thermostatic bath (measured by a 100 Ω platinum probe) was regulated around the set value. The relative uncertainty caused by the temperature regulator is 0.1%.

2.3. Measurement System. The parameters measured in this experiment included temperature, pressure, and mixture composition (CO₂ molar fraction).

Temperature was measured by 100 Ω platinum probes, calibrated by the CAS (Chinese Academy of Sciences) to ensure a temperature accuracy of ± 0.05 $^{\circ}\text{C}$. There were four temperature measurement points, three located one each at the bottom, middle, and top of the thermostatic bath to measure the vertical temperature gradient (verified within 0.01 $^{\circ}\text{C}/\text{cm}$); and the fourth in the middle of the equilibrium cell. The platinum probes in the middle of the thermostatic bath and the middle of equilibrium cell were at the same height.

The pressure in the equilibrium cell was measured by an external pressure transducer (PT) (NS-II, Shanghai TM Automation Instruments Co. Ltd., Shanghai, China) that could work at pressures up to 6 MPa, with a relative uncertainty of 0.1% FS (full scale), i.e. ± 6 kPa. Pressure has little effect on solubility,^{9,13,14} so the accuracy of pressure measurement has little effect on the experimental results.

The sample composition was analyzed by a gas chromatograph (GC1690, Hangzhou Kexiao Chemical Equipment Co., Ltd., Hangzhou, China) equipped with both a thermal conductivity detector (TCD) and a flame ionization detector (FID). The former was used when the CO_2 molar fraction was higher than 1000 ppm, while the latter was used when the CO_2 molar fraction was below 1000 ppm. The minimum detection limit was 1 ppm and the relative uncertainty was 0.1% for the TCD and 0.5% for the FID. The area external standard method was used for analysis, so standard gases were required to generate a methods file. For each composition of solvent, standard gases with similar composition and with different CO_2 contents were calibrated and the corresponding method files were obtained. In this way, the relative uncertainty due to calibration uncertainties and analysis dispersion were estimated to be within 1%. Therefore, the total relative uncertainties introduced by the GC was 1.1% for the TCD and 1.6% for the FID.

Fifteen standard gas samples were used in this experiment, according to the estimated CO_2 solubility (provided by Nanjing Special Gas Co., Ltd., Nanjing, China), all of which were ternary mixtures of $\text{CH}_4\text{--N}_2\text{--CO}_2$, with the mole fractions of N_2 and CO_2 listed in Table 1 (with CH_4 as the remainder). The relative

Table 1. Molar Fractions of N_2 and CO_2 for Fifteen Standard Gas Samples

N_2	CO_2			
	0.001%	0.01%	0.1%	1%
0		1#	2#	3#
10%	4#	5#	6#	
30%	7#	8#	9#	
50%	10#	11#	12#	
70%	13#	14#	15#	

uncertainty of the purities was 1% of the labeled concentration, except that the absolute uncertainty was 1 ppm for CO_2 in the 4#, 7#, 10#, and 13# gases.

2.4. Chemicals. The CH_4/N_2 mixtures and CO_2 used in this work were prepared and provided by Shanghai Wetry Standard Gas Co., Ltd. (Shanghai, China). CO_2 and the chemicals used for mixture preparation had the following certified purities, checked by gas chromatography:

CO_2 : 99.995 mol %

CH_4 : 99.99 mol %

N_2 : 99.999 mol %

Because CO_2 was supplied in quantities at least double the estimated solubility, the impurities in CO_2 , and the CO_2 present

in CH_4 or N_2 feed gases, if any, would not result in insufficient or excessive solid formation of CO_2 . Thus, these purities had little effect on experimental results.

3. EXPERIMENTAL PROCEDURE

The experimental method used to measure the SLE data was the static analytic method with liquid phase sampling and analysis. The experimental procedure consisted of three steps: filling the equilibrium cell, setting up experimental conditions, and conducting the measurements at the equilibrium. In the following, more details are provided, referring to Figure 1.

3.1. Filling the Equilibrium Cell. (1) All of the solute (CO_2) and part of the solvent (CH_4/N_2 mixture) were introduced into the evacuated equilibrium cell (EC) at room temperature.

First, excess CO_2 had to be loaded completely at room temperature because it would freeze out at low temperatures. Subsequently, some solvent also had to be loaded at room temperature to dilute CO_2 in the tubes before it solidified.

The quantity of filled CO_2 was controlled by observing the pressure in the EC. The mass quantity of required CO_2 was first calculated based on the estimated solubility of CO_2 under the conditions studied (solvent composition and given temperature). The amount used was several times the soluble quantity, and the density of CO_2 in the cell was then able to be calculated. Finally, the pressure in EC after completing the CO_2 filling procedure at room temperature could be estimated, at about 300–500 kPa when the working temperature was lower than -150 $^{\circ}\text{C}$, and this increased with temperature. When the temperature reached -100 $^{\circ}\text{C}$, the pressure was about 3000 kPa.

(2) After loading CO_2 and the solvent until the pressure of the EC reached about 5 MPa, the thermostatic bath (containing the EC) was placed into the cryostat, which was partially filled with liquid nitrogen.

(3) As the temperature and consequently the pressure in the EC decreased, more solvent was loaded to ensure that the quantity introduced was sufficient to fill 80% of the EC with the liquid phase. The quantity of solvent could be controlled by observing the pressure drop in the solvent reservoir (a 40-L, 10-MPa steel cylinder), estimated to be about 0.2–0.4 MPa.

3.2. Setting up Experimental Conditions. After completing the filling process, the temperature was set as desired. The heating wires were activated and the heating power was varied under the control of the PID temperature regulator (first 0 and then 100% power and finally at a relatively stable percentage).

Meanwhile, helium was added into the bath to improve heat transfer and consequently accelerate equilibrium.

3.3. Conducting Measurements at Equilibrium. When equilibrium was reached (approximately 12 h after the change from room temperature to cryogenic temperature and approximately 4 h for each 5 $^{\circ}\text{C}$ decrease), the equipment was ready for sampling.

- (1) V5 was opened quickly and a small amount of saturated liquid sample was introduced into the evacuated buffer tank, where the liquid vaporized.
- (2) Valve V6 was then opened to introduce the vaporized sample into the gas chromatograph for analysis. For a given temperature, at least 10 samples were extracted and analyzed to check repeatability: the mean relative deviation of solubility due to the analysis dispersion was estimated to be within 3%.

After completing a data set for one solvent composition at a given temperature, another temperature was set and, while

waiting for equilibrium at this temperature, the buffer tank was evacuated. After completing a series of temperature points for this solvent composition, the thermostatic bath was removed from the cryostat and rewarmed to room temperature. The liquid in the equilibrium cell was vaporized and gently vented. After the temperature increased to room temperature, the equilibrium cell was evacuated and was ready for the next series of experiments using another solvent composition.

4. RESULTS AND DISCUSSION

In this section, data for CO₂ solubility in pure liquid methane were obtained using the apparatus and compared with both experimental data from Davis et al.⁷ and calculated data from Shen et al.⁸ to verify the accuracy of the approach. Subsequently, CO₂ solubility in CH₄/N₂ mixtures was measured at cryogenic temperatures and, adopting the PR EOS calculation method, CO₂ solubility was calculated and compared with the experimental results.

4.1. CO₂ Solubility in Pure CH₄. Considering the temperatures that may occur in an LNG process, seven temperature points in the range −103.3 to −143.5 °C from Davis et al.⁷ were chosen as the working temperatures, and two lower temperatures (−150 and −160 °C) were also measured.

The average values of ten samples were determined as the CO₂ solubility (x) and compared with the experimental data from Davis et al.⁷ (x_{exp}) listed in Table 2.

Table 2. Experimental Data for CO₂ Solubility in Pure CH₄ (Mole Fraction, 10³ ppm)^a

T (°C)	x (10 ³ ppm)	x_{exp} (10 ³ ppm)	Δ (%)
−160	0.172		
−150	0.823		
−143.5	1.413	1.6	−11.69
−137.9	2.479	2.5	−0.84
−133.7	3.678	3.7	−0.59
−128.6	5.665	5.8	−2.33
−122.8	8.225	9.3	−11.56
−111.1	17.64	18.3	−3.61
−103.3	28.96	29.0	−0.14
MRD			4.39

^a x : Experimental data from our work; x_{exp} : experimental data from Davis et al.⁷ $\Delta = 100(x - x_{\text{exp}})/x_{\text{exp}}$; MRD = $(100/N) \times \sum |(x - x_{\text{exp}})/x_{\text{exp}}|$.

Furthermore, calculation data from Shen et al.⁸ using EOS methods (PR and SRK) at corresponding temperatures were also compared with our experimental results, as shown in Figure 2.

It can be seen from Table 2 and Figure 2 that our experimental results were in good agreement with those of Davis et al.⁷ The deviations between the results were very small in most cases, except at two temperatures (−143.5 and −122.8 °C), with a satisfactory mean relative deviation of 4.39%. In fact, we can see from Figure 2 that our experimental results were a better match with the calculated results, and were more regular and smoother. On the contrary, the data of Davis et al.⁷ has an evident jump at −122.8 °C, which is not expected.

4.2. CO₂ Solubility in CH₄/N₂ Mixtures. The nitrogen content in CBM has a very wide range and the recommended specification for liquefaction is a level no higher than 70%. With this consideration, 4 compositions with nitrogen molar fractions of 10, 30, 50, and 70% were measured in this work. Furthermore, based on our former studies into CBM liquefaction processes,^{16–18} the lowest temperature in a liquefaction process

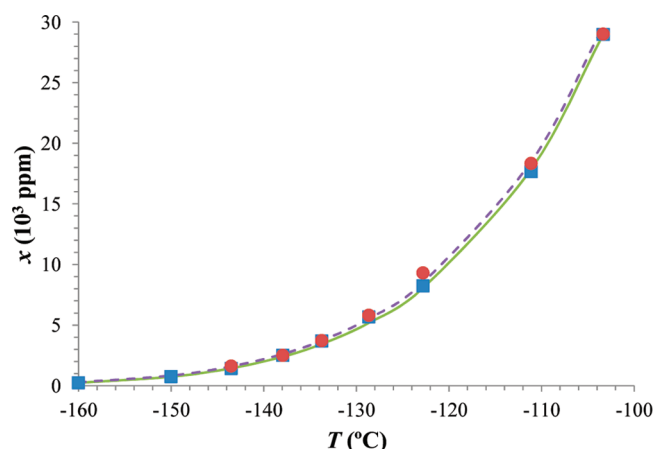


Figure 2. Solubility of CO₂ in saturated liquid methane (mole fraction). (■) Experimental data (this work); (●) Davis et al.;⁷ (—) Shen et al.⁸ (PR); (---) Shen et al.⁸ (SRK).

can vary with the feed composition and the storage pressure, decreasing when the feed nitrogen content increases or as storage pressure decreases, with a resulting range of about −150 to −190 °C. As a result, the temperature points studied for each solvent composition were from −150 °C to the boiling point near atmospheric pressure, at 5 °C intervals.

The PR EOS method is widely used to model natural gas processing systems, so it was adopted for solid–liquid equilibrium calculations in this paper, as shown below.

To perform solid–fluid phase equilibrium for a ternary fluid phase (species 1, 2, and 3) with a pure solid phase (species 2), one solves the fugacity balance for species 2 between the phases:

$$f_2^L(T, P, x_2) = f_2^S(T, P) \quad (1)$$

where subscript 1 refers to the solvent CH₄, subscript 2 refers to the solute CO₂, subscript 3 refers to the solvent N₂, and f^L and f^S are the fugacity of pure solid and liquid, respectively.

According to the definition of the fugacity coefficient, eq 1 becomes

$$\phi_2^L P x_2 = \phi_2^S P_2^S \exp\left(\frac{V_2(P - P_2^S)}{RT}\right) \quad (2)$$

where x_2 is the mole fraction of CO₂ in the liquid phase, ϕ_2^L is the liquid phase partial fugacity coefficient for CO₂, P is the system pressure in Pa, P_2^S is the saturated vapor pressure of solid CO₂ at the system temperature, in Pa, ϕ_2^S is the fugacity coefficient of pure CO₂ at the system temperature, and V_2 is the molar volume of solid CO₂.

By simplification, we can obtain:

$$x_2 = \frac{\phi_2^S P_2^S \exp(V_2(P - P_2^S)/RT)}{\phi_2^L P} \quad (3)$$

where the fugacity coefficient ϕ can be evaluated using the standard form of the PR EOS,⁹ which can be written as eq 4 or eq 5

$$P = \frac{RT}{V_m - b} - \frac{a}{V_m(V_m + b) + b(V_m - b)} \quad (4)$$

$$z^3 - (1 - B)z^2 + (A - 3B^2 - 2B)z - (AB - B^2 - B^3) = 0 \quad (5)$$

where $A = (aP/(R^2T^2))$, and $B = (bP/RT)$.

Classical mixing rules can be used to calculate the values of a and b of the mixtures:

$$a = \sum_{i=1}^N \sum_{j=1}^N x_i x_j (a_i a_j)^{0.5} (1 - k_{ij})$$

$$= x_1^2 a_1 + x_2^2 a_2 + x_3^2 a_3 + 2x_1 x_2 (a_1 a_2)^{0.5} (1 - k_{12})$$

$$+ 2x_1 x_3 (a_1 a_3)^{0.5} (1 - k_{13}) + 2x_2 x_3 (a_2 a_3)^{0.5} (1 - k_{23}) \quad (6)$$

$$b = \sum_{i=1}^N x_i b_i = x_1 b_1 + x_2 b_2 + x_3 b_3 \quad (7)$$

where $a_i = 0.45724(R^2 T_{ci}^2)/P_{ci}[1 + m_i(1 + (T/T_{ci})^{0.5})]^2$, $b_i = 0.0778(RT_{ci})/P_{ci}$, and $m_i = 0.37464 + 1.5422\omega_i - 0.26992\omega_i^2$.

The mole fraction of component i is represented by x_i , and ω_i is the eccentric factor of component i . k_{ij} is the binary interaction parameter characterizing molecular interactions between molecules i and j , listed in Table 3. k_{ij} between CH₄ and CO₂ is derived

Table 3. k_{ij} Between CH₄, CO₂, and N₂ by the PR EOS

k_{ij}	CH ₄	CO ₂	N ₂
CH ₄		$0.0998 + 5.4835/T - 36.134/T^2$	0.036
CO ₂	$0.0998 + 5.4835/T - 36.134/T^2$		-0.02
N ₂	0.036	-0.02	

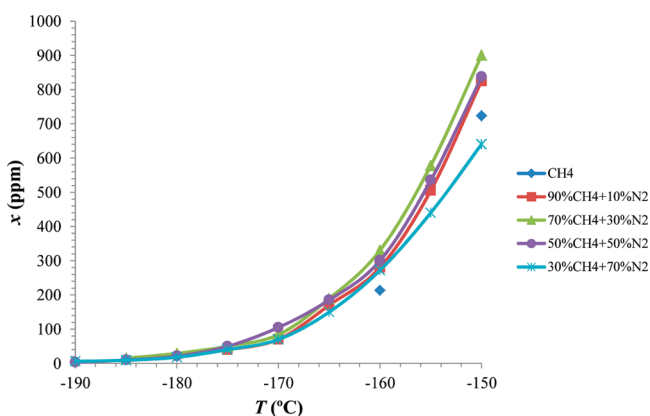


Figure 3. Solubility of CO₂ in saturated CH₄/N₂ mixtures (mole fraction).

from ZareNezhad et al.,¹⁹ and the rest of the data come from the software package ASPEN-HYSYS.

Using the PR equation of state, the fugacity coefficient of a component in the mixture can be evaluated by the following equation:

$$\phi_2^l = \exp \left(\frac{b_2}{b} (z - 1) - \ln(z - B) - \frac{A}{2\sqrt{2}B} \left(\frac{\bar{a}_2}{a} - \frac{b_2}{b} \right) \right.$$

$$\left. \times \ln \left(\frac{z + (\sqrt{2} + 1)B}{z - (\sqrt{2} - 1)B} \right) \right) \quad (8)$$

where $\bar{a}_2 = 2x_1 a_{12} + 2x_2 a_2 + 2x_3 a_{23} = 2x_2 a_2 + 2x_1 (a_1 a_2)^{0.5} (1 - k_{12}) + 2x_3 (a_2 a_3)^{0.5} (1 - k_{23})$.

Experimental results for CO₂ solubility in CH₄/N₂ mixtures (x) are shown in Figure 3, and are compared with those of the calculated data (x_{cal}), as listed in Table 4. Furthermore, for comparison, parts of the experimental data for CO₂ solubility in pure methane are also plotted in Figure 3.

From Figure 3 it is evident that both temperature and nitrogen content have a great effect on CO₂ solubility: temperature has a positive effect (the higher the temperature, the larger the solubility), while the effect of nitrogen content is not monotonic (CO₂ solubility first increases and then decreases when the nitrogen content increases).

Table 4 shows a larger deviation between the calculated and experimental results when nitrogen content increases. This is because, in this PR EOS calculation model, the binary interaction coefficients between CH₄ and N₂, as well as that between N₂ and CO₂, were obtained from the HYSYS software, which were constant and inaccurate. As a result, when the nitrogen content increases, the negative effect of these inaccurate binary interaction coefficients rises and this greatly influences the accuracy of this calculation.

According to our experimental data, the CO₂ solubility changes only slightly with nitrogen content at a given temperature, with a slight increase when the nitrogen content increases. However, CO₂ solubility decreases rapidly with temperature. When the nitrogen content in the CBM increases, the liquefaction temperature decreases sharply due to the lower boiling point of nitrogen and the CO₂ solubility is consequently reduced to a very low value when liquefied CBM (LCBM) is stored at atmospheric pressure. For example, when the nitrogen content is 50%, the liquefaction temperature is about -190 °C when a LCBM is stored at 110 kPa, and the CO₂ solubility is only 3 ppm. In comparison, with CO₂ solubility in ordinary LNG is about 200 ppm (with a liquefaction temperature of around -160 °C), so the traditional CO₂

Table 4. Solubility of CO₂ in CH₄/N₂ Mixtures (Mole Fraction, ppm)^a

T (°C)	90%CH ₄ + 10%N ₂			70%CH ₄ + 30%N ₂			50%CH ₄ + 50%N ₂			30%CH ₄ + 70%N ₂		
	p (kPa)	x	x_{cal}	p (kPa)	x	x_{cal}	p (kPa)	x	x_{cal}	p (kPa)	x	x_{cal}
-150	571	825	830	1115	900	930	1596	837	922	2023	640	756
-155	452	505	493	919	578	576	1270	535	599	1612	440	526
-160	347	280	280	715	330	342	1000	300	374	1240	272	350
-165	241	170	151	545	189	193	760	185	222	945	150	222
-170	175	70	77	413	84	104	564	105	126	703	70	134
-175	139	40	37	289	50	52	419	50	67	505	40	76
-180				200	29	24	292	21	34	353	18	41
-185				130	15	10	196	10	15	240	9	20
-190							124	3	7	145	6	9
MRD			5.39			10.39			39.22			67.40

^a x : Experimental data from our work; x_{cal} : calculated data using the PR EOS method; MRD = $(100/N) \times \sum |(x_{cal} - x)/x|$.

purification index of 50 ppm is clearly not applicable. Under these conditions, more complex CO₂ purification processes may be required and this would increase costs. One option is to raise the storage pressure and in this way, liquefaction temperature would be increased, along with CO₂ solubility. Thus, traditional CO₂ purification technologies may be directly applied.

5. CONCLUSION

A solid–liquid equilibrium experimental apparatus using a static-analytic method with gas chromatographic analysis was designed, constructed, and tested. Data for CO₂ solubility in pure liquid methane were first obtained and good agreement with experimental data available in the literature was confirmed, with a mean relative deviation of 4.39%. The solubility of CO₂ in CH₄/N₂ mixtures with nitrogen molar fractions of 10, 30, 50 and 70% was then measured at temperatures between –190 and –150 °C. This showed that CO₂ solubility changes only slightly with nitrogen content at a given temperature, initially increasing and then decreasing as the nitrogen content increases. However, CO₂ solubility decreases sharply with temperature and the liquefaction temperature decreases markedly as the nitrogen content in CBM increases. This reduces CO₂ solubility to a very low value when LCBM is stored at atmospheric pressure and may increase the costs for CO₂ purification. As a result, it is recommended that LCBM is kept at a higher pressure so that traditional CO₂ purification technologies can be directly applied for CBM liquefaction.

AUTHOR INFORMATION

Corresponding Author

*Tel.: +86-13621908234. Fax: +86-21-34206814. E-mail: linwsh@sjtu.edu.cn.

Notes

The authors declare no competing financial interest.

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